1. Compute $A + B$ and $-4A + B$ up to 4 decimal places if
$$A = \begin{pmatrix} -1 & 4 & -4 \\ 2 & 0 & -6 \\ 5 & -1 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 0 & 1 \\ 1.5 & -1.5 & -3 \\ 0.5 & 4 & -5 \end{pmatrix}.$$

2. Compute $\frac{3}{2}A - B$ and $\pi A + 1.34B$ up to 3 decimal places if
$$A = \begin{pmatrix} -1 & -3 & -3 & 8 \\ 3 & 7 & 4 & -2 \\ -4.2 & 0 & 2 & -2 \\ 7.5 & -2 & 1 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} -3 & -6 & -3 & 4 \\ -5 & 2 & -4 & -7 \\ 10 & 6 & 1 & -1 \\ -1 & 5 & -5 & 8 \end{pmatrix}.$$

3. Compute $A^T, B^T, (AB)^T$ and $(BA)^T$ up to 4 decimal places if
$$A = \begin{pmatrix} -1 & 4 & -4 \\ 2 & 0 & -6 \\ 5 & -1 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 0 & 1 \\ 1.5 & -1.5 & -3 \\ 0.5 & 4 & -5 \end{pmatrix}.$$

4. Compute $2A^T, 0.7B^T, \left(-\frac{3}{4}AB\right)^T$ and $\left(\sqrt{2.467}BA\right)^T$ up to 5 decimal places if
$$A = \begin{pmatrix} -1 & -3 & -3 & 8 \\ 3 & 7 & 4 & -2 \\ -4.2 & 0 & 2 & -2 \\ 7.5 & -2 & 1 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} -3 & -6 & -3 & 4 \\ -5 & 2 & -4 & -7 \\ 10 & 6 & 1 & -1 \\ -1 & 5 & -5 & 8 \end{pmatrix}.$$

5. Compute $A^T, \frac{B}{4.12}^T, (e^{-2}AB)^T$ and $\left(\frac{BA}{\sin 30}\right)^T$ up to 3 decimal places if
$$A = \begin{pmatrix} 0.912 & 4.322 & 3.876 \\ 4.560 & -1.009 & 2.871 \\ -7.238 & 5.557 & 2.226 \\ 1.211 & 7.001 & 0.999 \\ 7.912 & 5.677 & -8.933 \end{pmatrix} \text{ and } B = \begin{pmatrix} -10 & 0 & 3 & 2 & 5 \\ 9 & -3 & 1 & 8 & 5 \\ -6 & -2 & 1 & 3 & 8 \end{pmatrix}.$$

6. Compute $det(A), det(AB)$ and $det(-2.3BA)$ up to 6 decimal places if
$$A = \begin{pmatrix} -1 & 4 & -4 \\ 2 & 0 & -6 \\ 5 & -1 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 0 & 1 \\ 1.5 & -1.5 & -3 \\ 0.5 & 4 & -5 \end{pmatrix}.$$

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7. Compute $det(A) - 1.5678, det\left(AB - \frac{2}{7}A\right)$ and $det(BA + 3.061AB)$ up to 6 decimal places if

$$A = \begin{pmatrix} -1 & -3 & -3 & 8 \\ 3 & 7 & 4 & -2 \\ -4.2 & 0 & 2 & -2 \\ 7.5 & -2 & 1 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} -3 & -6 & -3 & 4 \\ -5 & 2 & -4 & -7 \\ 10 & 6 & 1 & -1 \\ -1 & 5 & -5 & 8 \end{pmatrix}.$$

8. Compute $A^{-1}, B^{-1}, (-2AB)^{-1}$ and $(0.55BA)^{-1}$ up to 4 decimal places if

$$A = \begin{pmatrix} -1 & 4 & -4 \\ 2 & 0 & -6 \\ 5 & -1 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 0 & 1 \\ 1.5 & -1.5 & -3 \\ 0.5 & 4 & -5 \end{pmatrix}.$$

9. Compute $(A + 3B)^{-1}, 2\pi B^{-1}, (det(A) \times AB)^{-1}$ and $(det(A) \times det(B) \times BA)^{-1}$ using scientific number format if

$$A = \begin{pmatrix} -1 & -3 & -3 & 8 \\ 3 & 7 & 4 & -2 \\ -4.2 & 0 & 2 & -2 \\ 7.5 & -2 & 1 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} -3 & -6 & -3 & 4 \\ -5 & 2 & -4 & -7 \\ 10 & 6 & 1 & -1 \\ -1 & 5 & -5 & 8 \end{pmatrix}.$$

10. Compute $(7A - B)^{-1}, (det(A) \times AB)^{-1}$ and $\left(\left(det(A) - det(B)\right) \times BA\right)^{-1}$ up to 3 decimal places if

$$A = \begin{pmatrix} 9.87 & 1.25 & -0.81 & 3.25 & -0.35 \\ 2.34 & 1.39 & -4.32 & 2.98 & 4.37 \\ -0.96 & 4.66 & 1.22 & 4.53 & 3.49 \\ -0.12 & -3.72 & 1.50 & 3.97 & 3.25 \\ 4.32 & 2.01 & 7.44 & -0.87 & 6.10 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1.26 & -0.88 & 3.42 & 7.89 & 0.23 \\ 4.51 & 5.12 & 2.35 & 5.34 & 0.38 \\ 4.33 & 4.12 & -3.22 & 0.07 & 0.99 \\ 2.90 & 3.22 & 1.24 & -3.67 & 1.78 \\ -6.03 & -9.32 & 1.88 & -7.44 & 5.96 \end{pmatrix}.$$

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